

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

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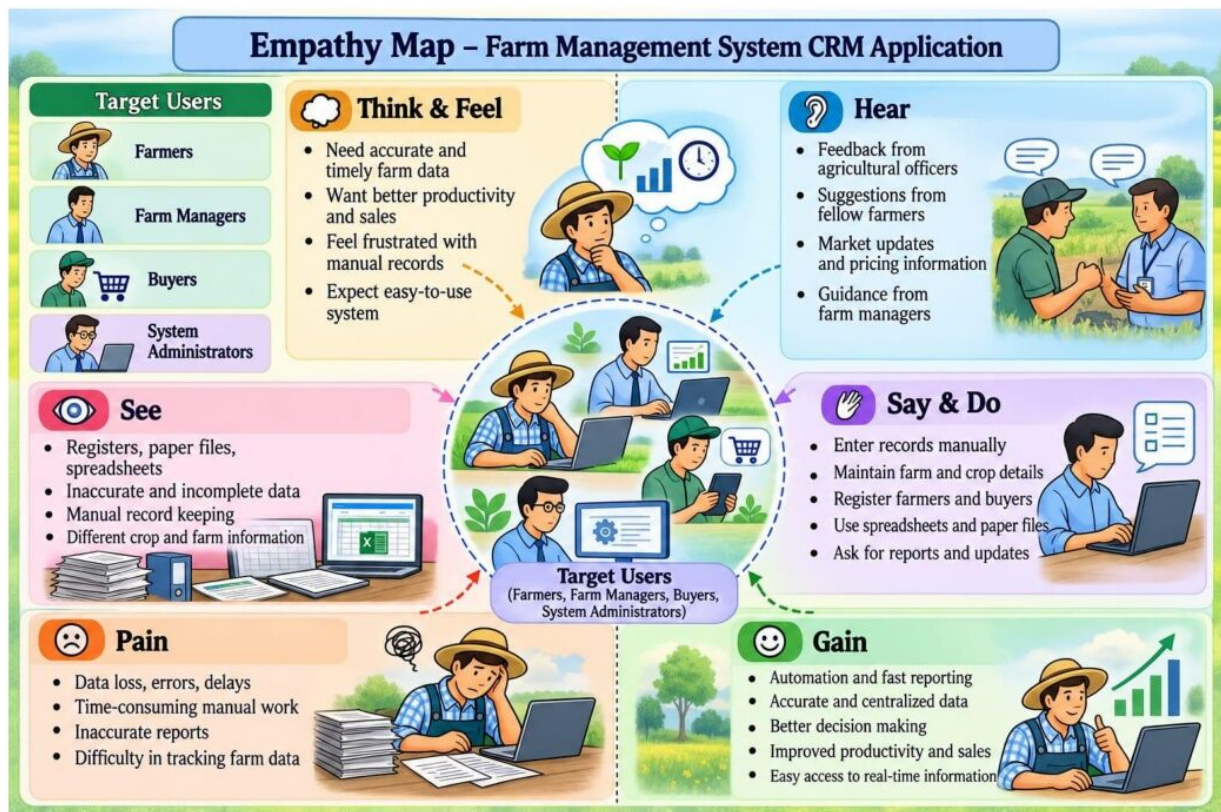
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



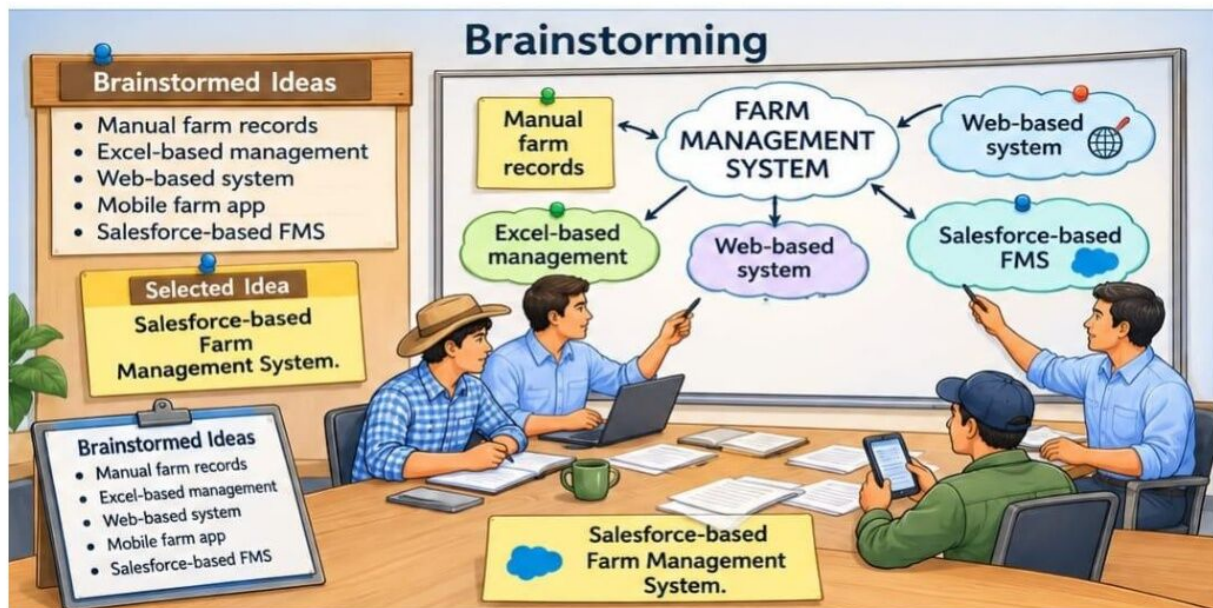
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

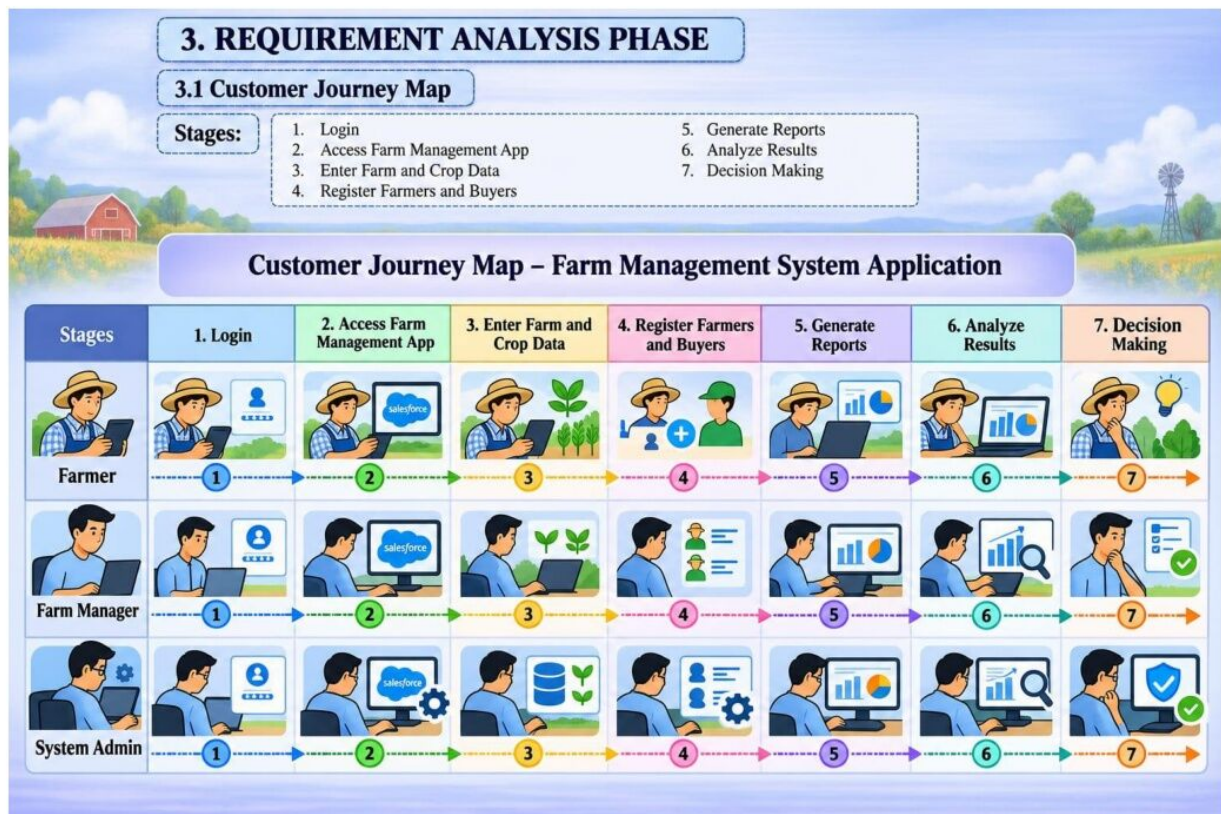


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6	Dashboard Visualization
FR-7	User Role Management

Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards

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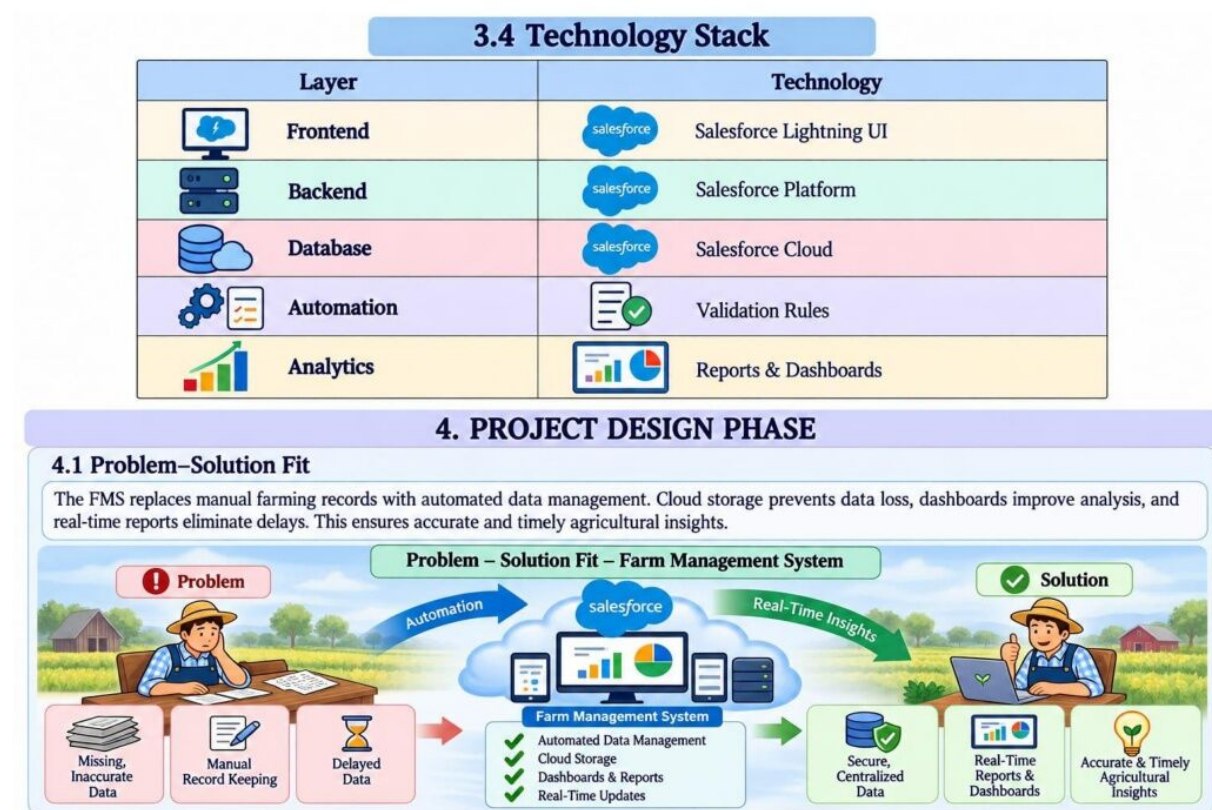
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

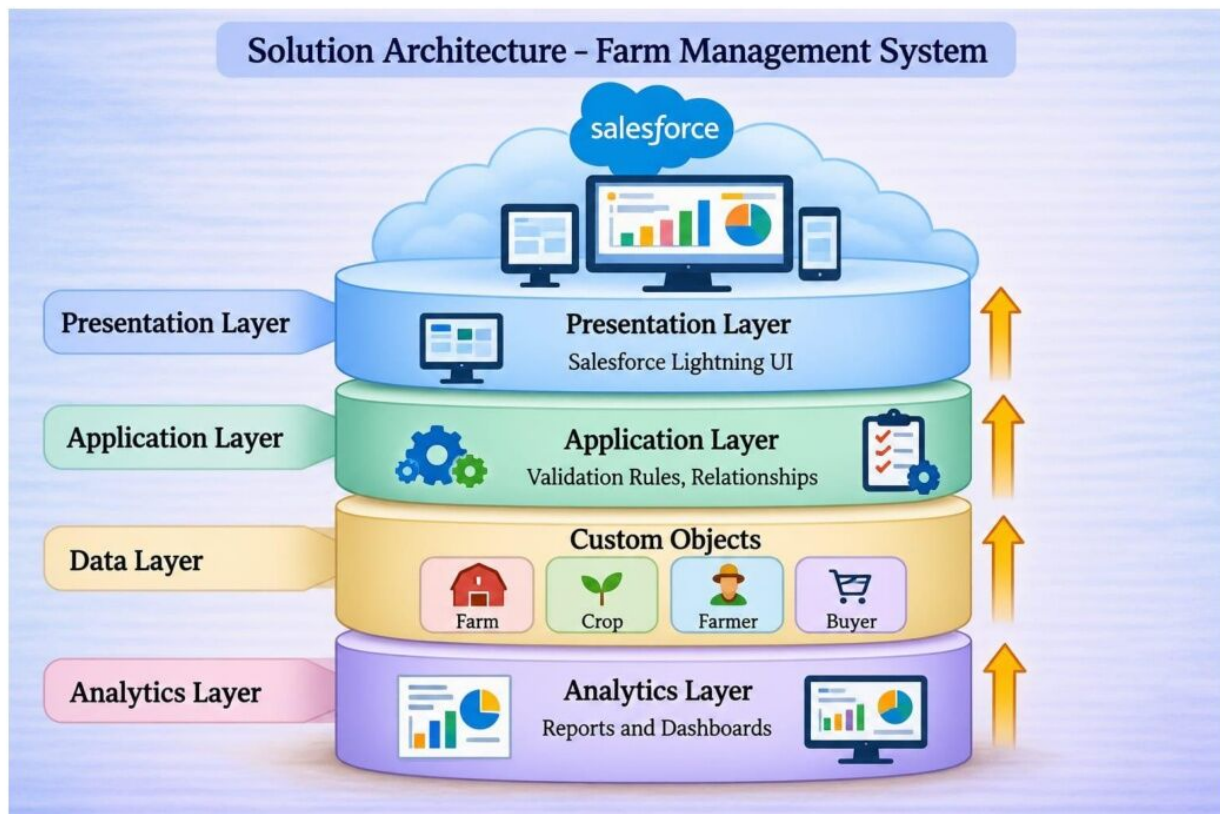
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

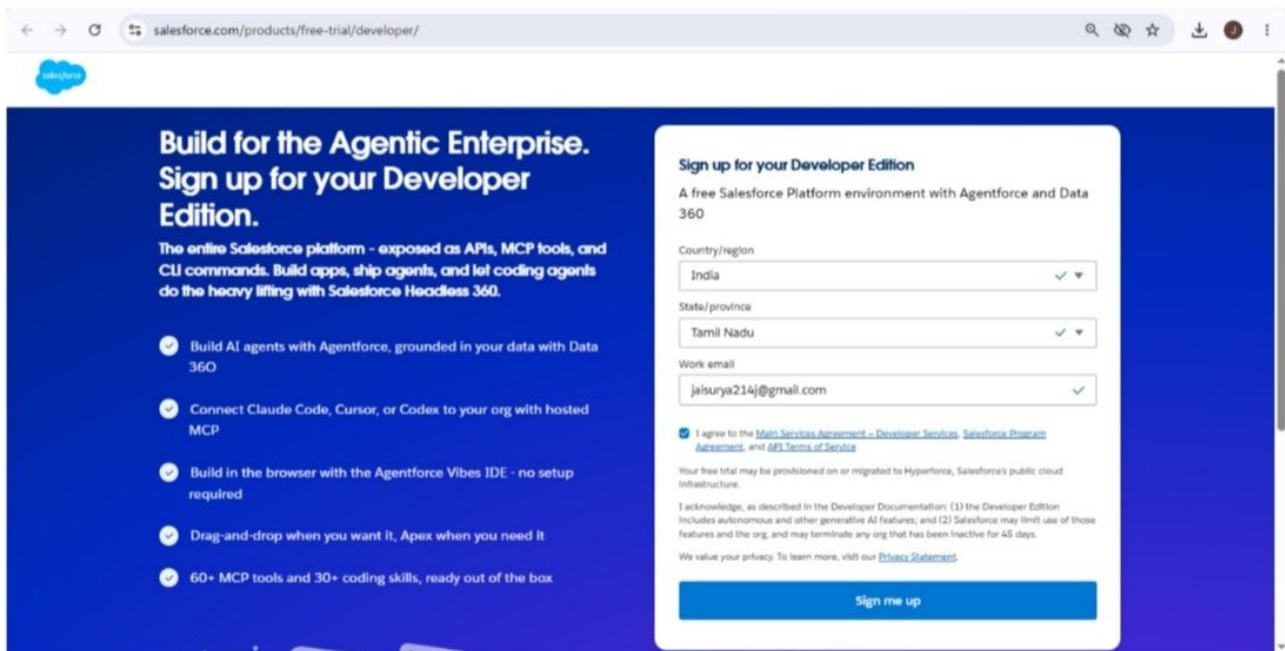
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

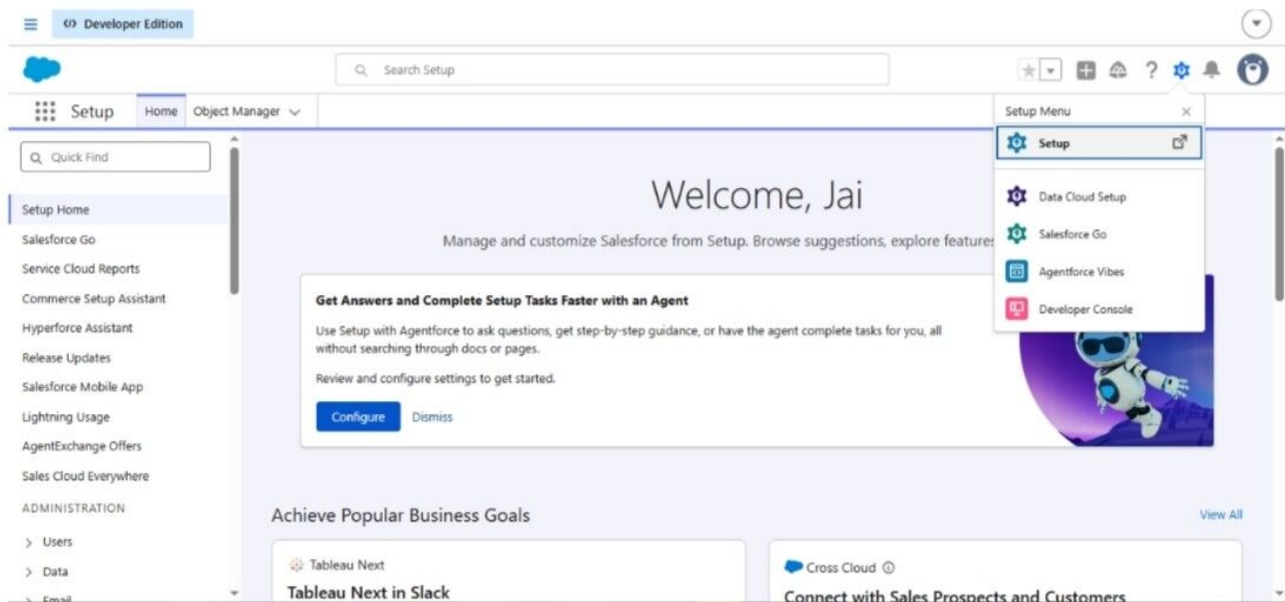
1. Search Developer.salesforce.com/signup
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



The screenshot shows the Salesforce Developer Edition sign-up page. The page has a dark blue header with the Salesforce logo. The main content area is white and contains the following elements:

- Build for the Agentic Enterprise. Sign up for your Developer Edition.**
- The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.**
- Features:**
 - Build AI agents with Agentforce, grounded in your data with Data 360
 - Connect Claude Code, Cursor, or Codex to your org with hosted MCP
 - Build in the browser with the Agentforce Vibes IDE - no setup required
 - Drag-and-drop when you want it, Apex when you need it
 - 60+ MCP tools and 30+ coding skills, ready out of the box
- Sign up for your Developer Edition**
 - A free Salesforce Platform environment with Agentforce and Data 360
 - Country/region: India (dropdown)
 - State/province: Tamil Nadu (dropdown)
 - Work email: jalsurya214@gmail.com (dropdown)
 - ☒ I agree to the [Main Services Agreement - Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)
 - Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.
 - I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.
 - We value your privacy. To learn more, visit our [Privacy Statement](#).
 - Sign me up** (button)

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

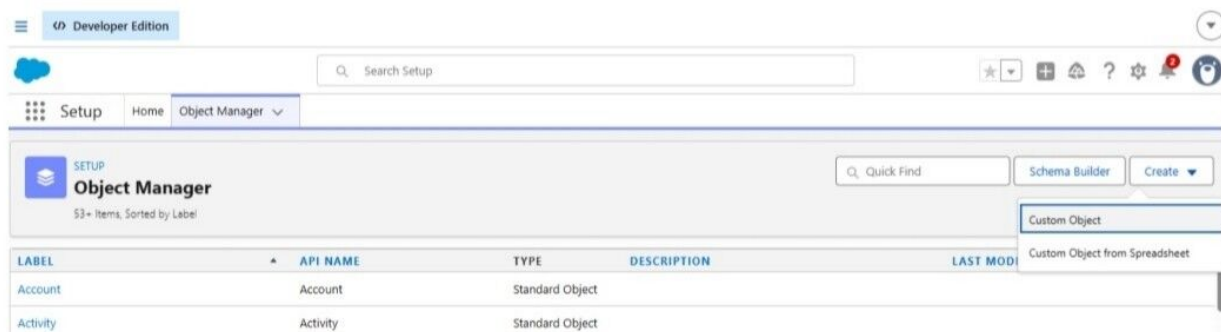
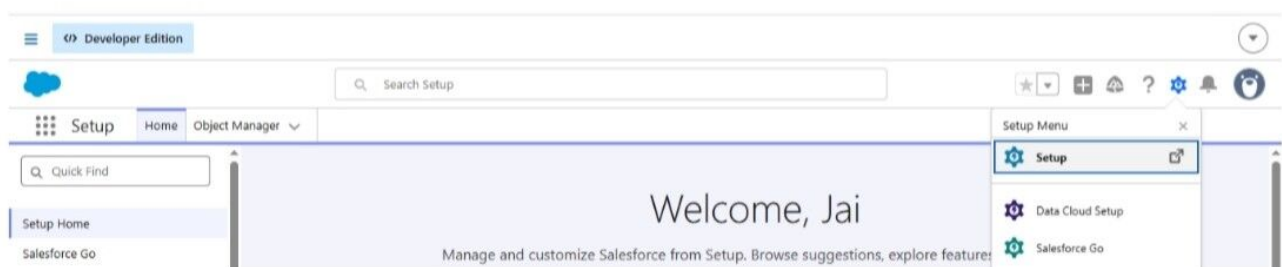
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

The screenshot shows the 'Details' tab of the Salesforce Object Settings page for an object named 'Farm'. The 'Data Type' is set to 'Text', with a warning: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' The left sidebar lists various configuration areas: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is divided into several sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Activities' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked); 'Object Classification' with a note about Enterprise vs. Light Application objects and checkboxes for 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (unchecked) and 'Deployed' (checked); and 'Search Status' with a note about searchability and a checked 'Allow Search' checkbox. At the bottom right are buttons for 'Save', 'Save & New', and 'Cancel'.

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Data Type: **Text** Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [...](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#).

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status [What is this?](#)

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#).

- ☒ Allow Search

[Save](#) [Save & New](#) [Cancel](#)

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

- 1. Click on the gear icon and then select Setup.
- 2. Click on the object manager tab just beside the home tab.
- 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Milestone 3 -

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

- 1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

Salesforce Setup interface showing the **Custom Tabs** configuration page.

Left Navigation:

- Setup
- Home
- Object Manager
- Search: Tabs
- User Interface
- Rename Tabs and Labels
- Tabs**

Main Content Area:

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Action	Label	Tab style	Description
Edit / Del	Birds	Bird	
Edit / Del	Crop	Crop	
Edit / Del	Computer	Computer	
Edit / Del	Farmers	Farmer	
Edit / Del	Apple	Apple	
Edit / Del	Farms	Farm	
Edit / Del	Global	Global	

Web Tabs [New](#) [What is This?](#)

No Web Tabs have been defined

Visualforce Tabs [New](#) [What is This?](#)

No Visualforce Tabs have been defined

Lightning Component Tabs [New](#) [What is This?](#)

2. Under custom object tabs, click New.

Setup Home Object Manager

Search Setup

Search Tabs

User Interface

Rename Tabs and Labels

Tabs

Didn't find what you're looking for? Try using Global Search.

New Custom Object Tab

Step 1 of 3

Step 1. Enter the Details

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab

Select an existing custom object or [create a new custom object now](#).

Object:

Tab Style:

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link:

Enter a short description.

Description:

Next Cancel

3. For Object, select Farm.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app

The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains a navigation menu with 'Apps' expanded, showing 'App Manager', 'External Client Apps', and 'Lightning Out 2.0 Apps'. The main content area displays a table of 31 items, sorted by App Name. The table columns are App Name, Developer Name, Description, Last Modified Date, App Type, and Visible. The table lists various Salesforce apps, including Agentforce Grid, Agentforce Studio, All Tabs, Analytics Studio, App Launcher, Approvals, Automation, Bolt Solutions, Community, Content, Data Cloud, Data Manager, Developer Edition, and Digital Experiences.

	App Name	Developer Name	Description	Last Modified Date	App Type	Visible
1	Agentforce Grid	AIWorkbench	Access Agentforce Grid	8/3/2026, 11:01 AM	Lightning	✓
2	Agentforce Studio	AgentforceStudio	Agentforce Studio	8/3/2026, 11:01 AM	Lightning	✓
3	All Tabs	AllTabSet		8/3/2026, 10:54 AM	Classic	✓
4	Analytics Studio	Insights	Build CRM Analytics dashboards and apps	8/3/2026, 10:54 AM	Classic	✓
5	App Launcher	AppLauncher	App Launcher tabs	8/3/2026, 10:54 AM	Classic	✓
6	Approvals	Approvals	Manage approvals and approval flows	8/3/2026, 10:54 AM	Lightning	✓
7	Automation	FlowsApp	Automate business processes and repetitive tasks.	8/3/2026, 11:00 AM	Lightning	✓
8	Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	8/3/2026, 10:54 AM	Lightning	✓
9	Community	Community	Salesforce CRM Communities	8/3/2026, 10:54 AM	Classic	✓
10	Content	Content	Salesforce CRM Content	8/3/2026, 10:54 AM	Classic	✓
11	Data Cloud	Audience360	Build a thorough and complete understanding of your customers.	8/3/2026, 10:54 AM	Lightning	✓
12	Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes.	8/3/2026, 10:54 AM	Lightning	✓
13	Developer Edition	Developer_Edition	Welcome to your Developer Edition Org.	8/3/2026, 11:11 AM	Lightning (Managed)	✓
14	Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	8/3/2026, 10:54 AM	Lightning	✓

- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
 - From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name [?]
Name your app...

* Developer Name [?]
Enter a developer name...

Description [?]
Enter a description...

App Branding

Image [?]
 Upload

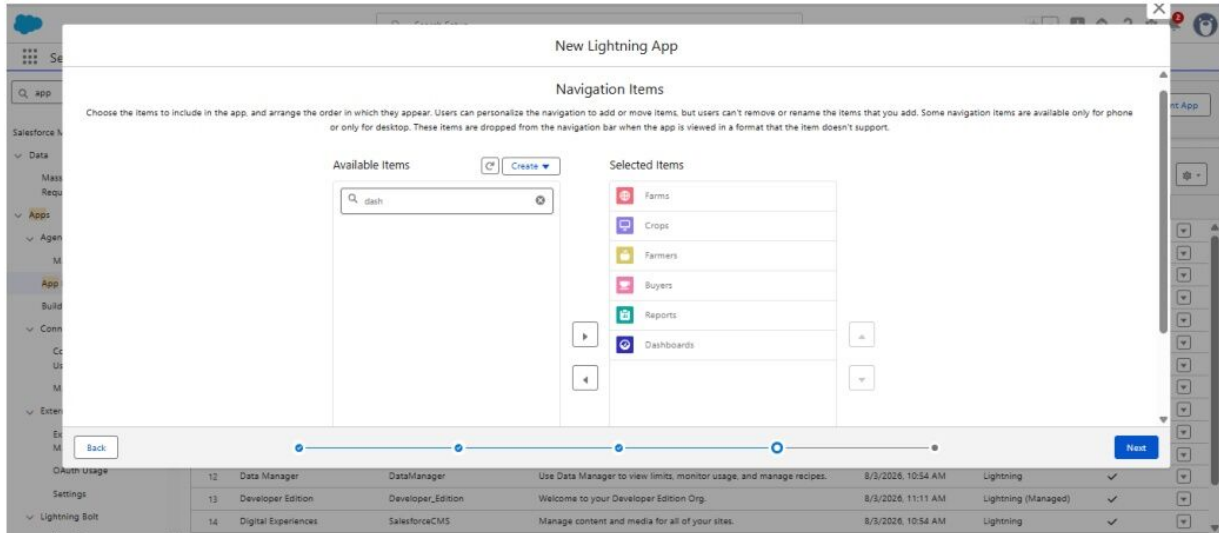
Primary Color Hex Value [?]
#0070D2

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview

12	Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes.	8/3/2026, 10:54 AM	Lightning	✓
13	Developer Edition	Developer_Edition	Welcome to your Developer Edition Org.	8/3/2026, 11:11 AM	Lightning (Managed)	✓
14	Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	8/3/2026, 10:54 AM	Lightning	✓

3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

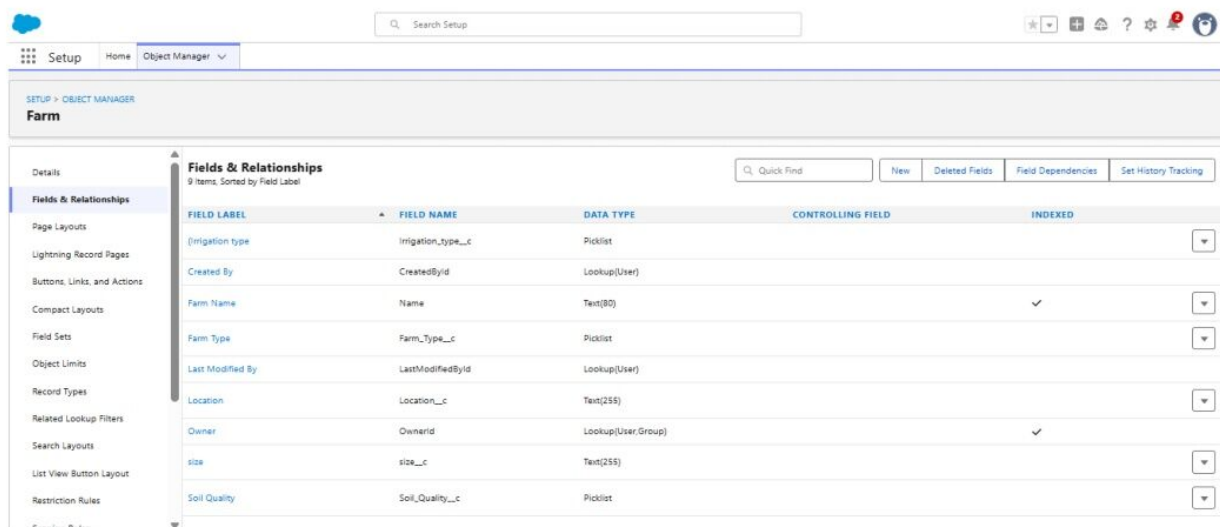
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



The screenshot shows the Salesforce Setup interface for the 'Farm' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Fields & Relationships' and shows a list of fields for the 'Farm' object. The fields are sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are: Irrigation type (Picklist), Created By (Lookup(User)), Farm Name (Text(80)), Farm Type (Picklist), Last Modified By (Lookup(User)), Location (Text(255)), Owner (Lookup(User,Group)), size (Text(255)), and Soil Quality (Picklist).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Irrigation type	Irrigation_type__c	Picklist		
Created By	CreatedByid	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Type	Farm_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	Text(255)		
Owner	OwnerId	Lookup(User,Group)		✓
size	size__c	Text(255)		
Soil Quality	Soil_Quality__c	Picklist		

5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface. At the top, there is a navigation bar with 'Setup', 'Home', and 'Object Manager'. A search bar is also present. The main content area is titled 'Farm' and shows the 'Fields & Relationships' section. On the left, a sidebar lists various setup options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main area displays a list of data types for the 'Farm' object. The 'Picklist' option is selected, and its description is shown on the right. The description states: 'Allows users to select a value from a list you define.' Below this, there are several other data types listed, each with a brief description of its functionality.

Data Type	Description
☐ Master-Detail Relationship	Create a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship field is required on all detail records. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create roll-up summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
☐ External Lookup Relationship	Create a relationship that links this object to an external object whose data is stored outside the Salesforce org.
☐ Checkbox	Allows users to select a True (checked) or False (unchecked) value.
☐ Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
☐ Date	Allows users to enter a date or pick a date from a popup calendar.
☐ DateTime	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the DateTime field.
☐ Email	Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
☐ Geolocation	Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distances.
☐ Number	Allows users to enter any number. Leading zeros are removed.
☐ Percent	Allows users to enter a percentage number; for example, "10" and automatically adds the percent sign to the number.
☐ Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
☒ Picklist	Allows users to select a value from a list you define.
☐ Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
☐ Text	Allows users to enter any combination of letters and numbers.
☐ Text Area	Allows users to enter up to 255 characters on separate lines.
☐ Text (Long)	Allows users to enter up to 131,072 characters on separate lines.

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save.

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, and Restriction Rules. The main content area is titled 'Step 2. Enter the details' and 'Step 2 of 4'. It contains the following fields and options:

- Field Label:** Farm Type
- Values:**
 - ☐ Use global picklist value set
 - ☒ Enter values, with each value separated by a new line
- Values List:** Crop, Livestock, Mixed
- ☐ Display values alphabetically, not in the order entered
- ☐ Use first value as default value
- ☒ Restrict picklist to the values defined in the value set
- Field Name:** Farm_Type
- Description:**
- Help Text:**

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.



Search Setup



Setup Home Object Manager

Setup > OBJECT MANAGER

Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Step 1. Choose the field type

Step 1

Next Cancel

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

Select one of the data types below.

☐ Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☒ Formula

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.
- You can create roll-up summary fields on the master record to summarize the detail records.

The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

☐ External Lookup Relationship

Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.

☐ Checkbox

Allows users to select a True (checked) or False (unchecked) value.

☐ Currency

Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes the Salesforce logo, a search bar labeled 'Search Setup', and several utility icons. Below the navigation bar, the breadcrumb trail reads 'SETUP > OBJECT MANAGER' and the current page is titled 'Crop'. A left-hand sidebar contains a list of setup categories: Details, Fields & Relationships (highlighted), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, and Restriction Rules. The main content area is titled 'Step 3. Enter formula' and includes navigation buttons 'Previous', 'Next', and 'Cancel'. It contains instructions: 'Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.' Below this, an example formula is shown: 'Example: Gross Margin = Amount - Cost_c'. There are two tabs: 'Simple Formula' and 'Advanced Formula'. Under 'Simple Formula', there are dropdowns for 'Select Field Type' (set to 'Crop') and 'Insert Field' (set to 'Insert Merge Field'). A 'Use Formula Assistant' button is present. Under 'Advanced Formula', the formula 'Expected Yield (Currency) = Actual_Yield__c / Total_Acres_Harvested__c' is entered in a large text area. A 'Quick Tips' sidebar on the right lists 'Getting Started' and 'Operators & Functions'.

Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label(Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label(State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields&Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

New Field Dependency

Help for this Page

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling field	-None-	Continue Cancel
Dependent field	-None-	Continue Cancel

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Setup

Home

Object Manager

Search Setup

prof

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP

Profiles

Clone Profile

Help for this Page

Enter the name of the new profile.

You must select an existing profile to clone from.

Existing Profile

User License

Profile Name

Standard User

Salesforce

Sales Department

Save

Cancel

5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option

The screenshot displays the Salesforce Setup interface for the 'Roles' section. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'Roles' and includes a search bar. Below the search bar, there is a section titled 'Understanding Roles' which explains the role hierarchy and provides a sample hierarchy diagram. The diagram shows a hierarchy starting with 'Executive Staff' (CEO, President, CFO, VP, Sales) at the top, branching into 'Western Sales Director', 'Eastern Sales Director', and 'International Sales Director'. Each director role has associated 'Rep' (Representative) roles below them. The 'Western Sales Director' has 'Western Sales Rep' and 'US Sales Rep'. The 'Eastern Sales Director' has 'Eastern Sales Rep' and 'NA Sales Rep'. The 'International Sales Director' has 'Asian Sales Rep' and 'European Sales Rep'. To the right of the diagram, there are notes explaining the permissions for each level, such as 'View & edit data, roll up forecasts, & generate reports for all users below' for the Executive Staff level. At the bottom right, there is a 'Set up Roles' button and a checkbox for 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.

The screenshot displays the Salesforce Setup interface for the 'Roles' section. The left sidebar contains a navigation menu with 'Users' and 'Roles' (highlighted). Under 'Roles', there are links for 'Feature Settings', 'Sales' (with sub-links for 'Contact Roles on Contracts', 'Contact Roles on Opportunities', and 'Add Role'), 'Service' (with sub-links for 'Case Teams', 'Case Team Roles', and 'Contact Roles on Cases'), and a search bar. The main content area is titled 'Roles' and 'Creating the Role Hierarchy'. It includes a sub-header 'Your Organization's Role Hierarchy' and a 'Show in tree view' button. The hierarchy is shown as a tree structure: 'Government Arts And science College Arakkonam' (with 'Add Role' link) is the parent of 'CEO' (with 'Add Role' link). Below 'CEO' are links for 'Edit', 'Delete', and 'Duplicate'. A message at the bottom states: 'Didn't find what you're looking for? Try using Global Search.'

Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup page for Users. The left sidebar has a search bar with 'users' entered. The 'Users' link is highlighted. The main content area is titled 'All Users' and contains a table of users. The table has columns for Action, Full Name, Alias, Username, Role, Active, and Profile. The table lists several users, including System Administrator, Chatter Free User, Standard Platform User, and Minimum Access - Salesforce.

Action	Full Name	Alias	Username	Role	Active	Profile
☐ Edit	Arumugam, Thennmothi	the	thennmothi@arumugam.25705210439@aperforce.com		✓	System Administrator
☐ Edit	Chatter, Robert	Chatter	chatter.00d00000000000000000000000000000@chatter.salesforce.com		✓	Chatter Free User
☐ Edit	Department, Sales	sdna	sales00d00000000000000000000000000000@gmail.com		✓	Standard Platform User
☐ Edit	EPIC, OrgFarm	CEPIC	epic.d0d00000000000000000000000000000@orgfarm.salesforce.com		✓	System Administrator
☐ Edit	Manager, Farm	f.man	farm00d00000000000000000000000000000@gmail.com	Farm Manager	✓	Standard Platform User
☐ Edit	User, Integration	inteq	integration@00d00000000000000000000000000000@zennel.com		✓	Analytics Cloud Integration User
☐ Edit	User, Security	sec	insightsecurity@00d00000000000000000000000000000@zennel.com		✓	Analytics Cloud Security User
☐ Edit	user, test	testuser	testuserhoney@gmail.com		✓	Minimum Access - Salesforce

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Setup

Home

Object Manager

Search Setup

Setup

Home

Object Manager

shar

Security

Guest User Sharing Rule Access Report

Sharing Settings

Didn't find what you're looking for? Try using Global Search.

SETUP

Sharing Settings

You can use sharing rules only to grant wider access to data, not to restrict access.

Step 1: Rule Name

Label Crop

Rule Name Crop

Description

Step 2: Select your rule type

Rule Type

☐ Based on record owner

☒ Based on criteria

Step 3: Select which records to be shared

Criteria

Field	Operator	Value	
None	None		AND
None	None		AND
None	None		AND
None	None		AND

Add Filter Logic...

Additional Options

☒ Include records owned by users who can't have an assigned role

Step 4: Select the users to share with

Share with

Public Groups

-- Select One --

Step 5: Select the level of access for the users

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web application interface with a modal form titled "Edit Ravi". The form is used for editing a farmer record. The fields and their values are as follows:

Field	Value
* Farmer Name	Ravi
Master Detail relationship	officer
Crop	Ravi
Contact Number	9178390329
State	Tamilnadu

The form also displays the "Owner" as Jai Surya. At the bottom right of the form are three buttons: "Cancel", "Save & New", and "Save". A note "* = Required information" is located in the top right corner of the form area. The background shows a blurred view of the application's main interface, including a sidebar with "Service", "Home", and "Chatter" tabs, and a "Recently Viewed" section.

Milestone 11-Reports:

What are Reports?

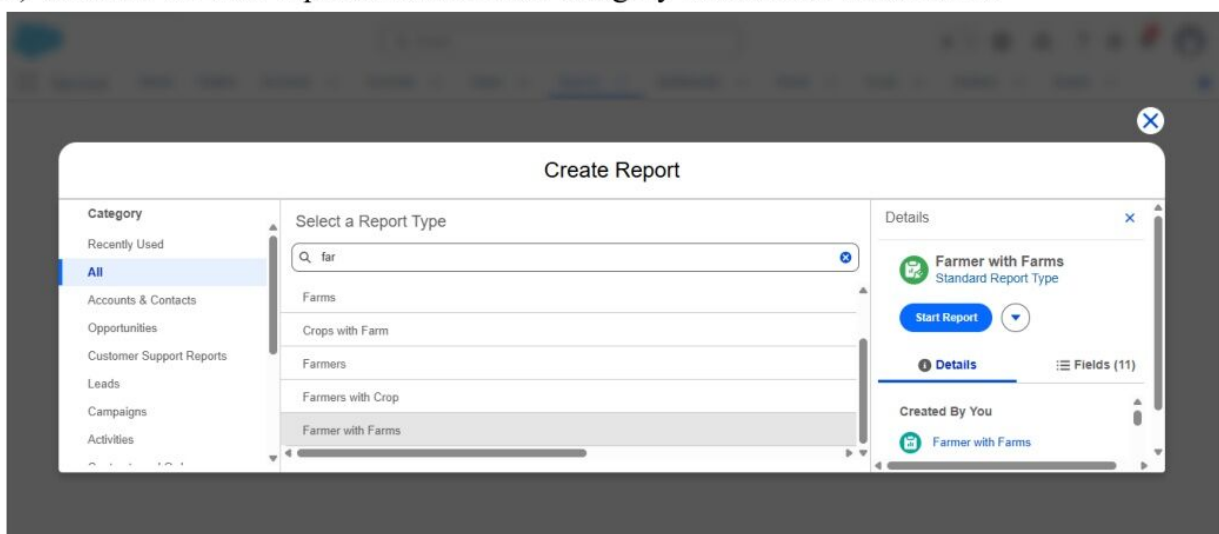
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot displays the Salesforce Reports interface. At the top, there is a navigation bar with a search bar and various utility icons. Below this is a menu bar with options like Service, Home, Chatter, Accounts, Contacts, Cases, Reports (selected), Dashboards, Farms, Crops, Farmers, and Buyers. The main section is titled 'REPORT' and shows the 'Farmer with Farms Report' with a sub-label 'Farmer with Farms'. Action buttons include 'Add Chart', 'Save & Run', 'Save', 'Close', and 'Run'. On the left, the 'Outline' tab is active, showing a 'Groups' section with an 'Add group...' button and a 'Columns' section with a list of fields: 'Farmer with Farm: ID', 'Farmer with Farm: Owner Name', 'Farmer with Farm: Owner Role', 'Farmer with Farm: Last Modified By', and 'Farmer with Farm: Created Date'. The main preview area shows a message: 'Previewing a limited number of records. Run the report to see everything.' and 'No records returned in preview. Try running the report or editing report filters.'

Activity 2:

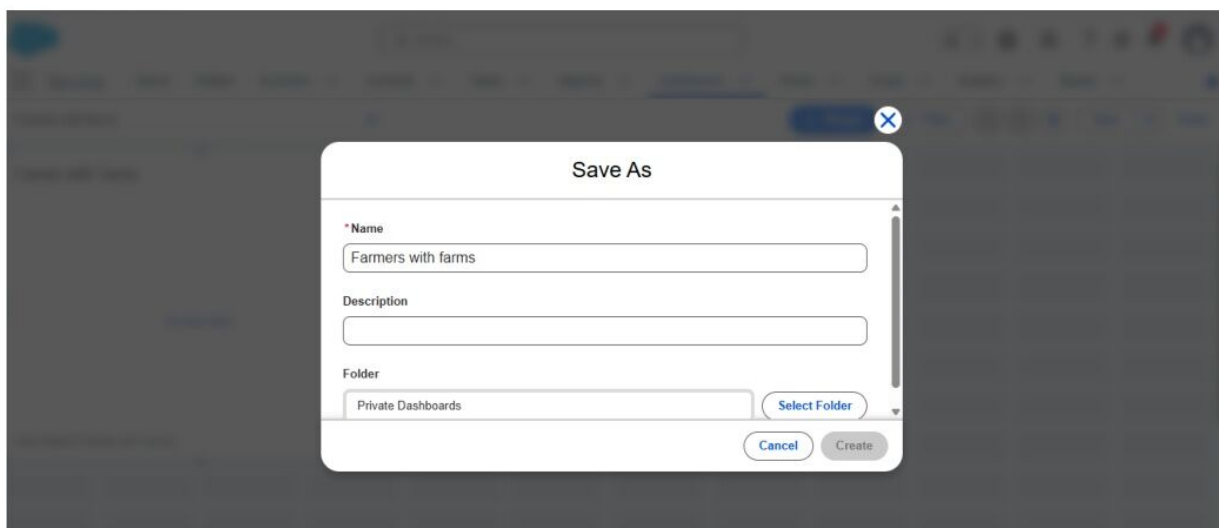
Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:**Create a Dashboard:**

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

Report: Farmers with Farms
Farmers with Farms Report

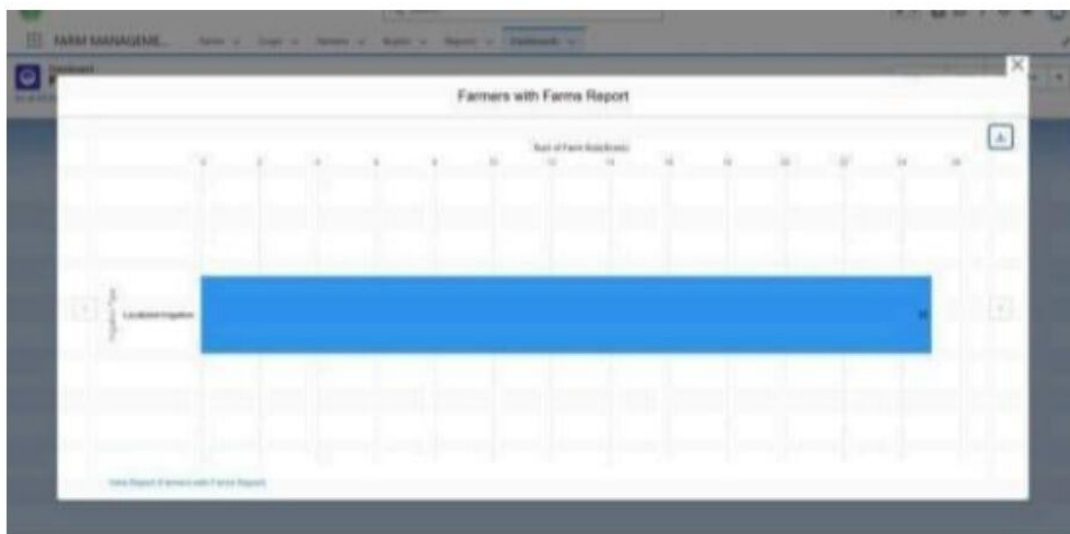
Total Records: 3
Total Farm Size (Hectares): 25

Location	Farmer Name	Farm Name	Contact Number	Farm Size (Hectares)	Farm Type
Rajesh	Rajesh	Rajesh Farm	9876543210	10	Cereals
Rajesh	Rajesh	Rajesh Farm	9876543210	10	Vegetables
Rajesh	Rajesh	Rajesh Farm	9876543210	10	Cereals
				25	
Total (3)				25	

Report Controls: [Filter] [Export] [Print] [Refresh] [Reset]

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You